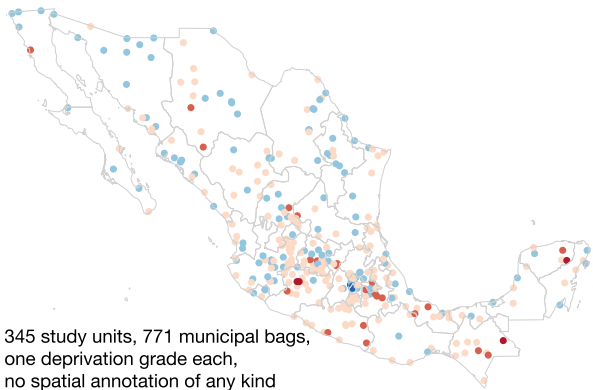
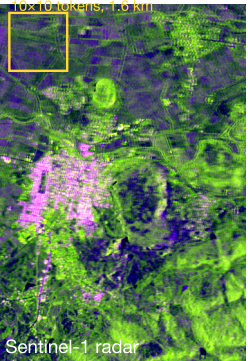


1 Published municipal aggregates



2 Free imagery, 160 m tokens



3 Three ways to learn from a bag label

Attention MIL (ABMIL, CLAM)

pool tokens, then predict the bag

map AUROC 0.46: chance

Label proportions (this work)

predict every token, average, match the share

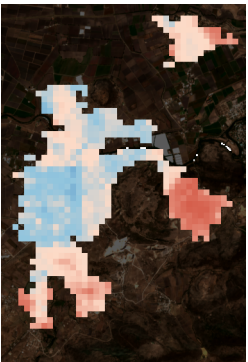
map AUROC 0.80 · within-municipality p 0.20 on test

Fully supervised oracle

same head, tract labels: the upper bound

within-municipality p 0.23 on test

4 Map, from aggregates alone



Tract truth, never trained on

